

ZETASTAB BR

HYDROGEN PEROXIDE STABILIZER / BLEACH REGULATOR

Product Description:

Zetastab BR is the indispensable peroxide stabilizer and oxidative bleach regulator. Specifically formulated to excel in oxidative bleaching baths utilizing hydrogen peroxide as the primary bleaching agent in the presence of strong alkalis at elevated temperatures, it functions as a negative catalyst. While it does not actively participate in the primary reaction, it plays a vital role in stabilizing the decomposition of hydrogen peroxide by inhibiting or slowing down the exothermic reaction.

The inclusion of sequestering agents in Zetastab BR effectively neutralizes metallic impurities by forming larger complexes, ensuring they remain dispersed within the bath. By preventing alkaline earth metal ions and heavy metal ions such as Ca^{2+} , Mg^{2+} , Cu^{2+} , Fe^{3+} , $\text{Mn}^{2+}/^{4+}$ from accelerating the bleaching reaction, it safeguards against rapid degradation of cellulose fibre polymer, preserving fabric strength. Unlike silicate-based peroxide stabilizers, Zetastab BR does not cause scaling on metallic machine parts or impart a harsh feel to the fabric.

Key Benefits:

- 1. Controlled Bleaching:** Slows down the decomposition rate of hydrogen peroxide in oxidative bleaching baths, ensuring a controlled reaction and achieving optimum whiteness.
- 2. Non-participation in Primary Reaction:** Does not actively engage in the primary bleaching reaction, preserving the integrity of the bleaching process.
- 3. Metal Ion Neutralization:** Neutralizes metallic ions by forming larger complexes, preventing them from accelerating uncontrolled peroxide decomposition and enhancing bleaching efficiency.
- 4. Silicone-Free:** Free from silicone compounds, resulting in a gentle feel on fabrics without any harshness — unlike conventional silicate-based stabilizers.
- 5. Scaling Prevention:** Prevents scaling on metallic machine parts, ensuring smooth operation and prolonging equipment lifespan.
- 6. Fabric Protection:** Guards against catalytic damage to fabrics during oxidative bleaching, maintaining fabric quality, strength, and durability.

Properties:

Chemical Constitution	Formulated with Na-salts of phosphonic acid and carbonic acid
Ionic Character	Anionic
pH of 1% solution	Approx. 4 to 5
Specific Gravity at 20°C	Approx. 1 gm/cc
Physical Form	Liquid
Storage Stability	Stable for 1 year at 20°C; sensitive above 50°C and below 0°C

Stability to Alkalis	Very Good
Stability to Acids	Very Good
Stability to Oxidants	Very Good

Applications:

1. Scouring & Oxidative Bleaching Bath — RFD / RFP Fabrics

Function	Product	Dosage	Parameters & Conditions
Lubricating Agent	Zetalub EAC	0.5%–1.0%	Auxiliary dosing – raise to 60°C – Caustic Soda dosing (2 cycles) – H ₂ O ₂ 50% dosing (2 cycles) – raise to optimum temp & hold (95°C × 45–60 min or 110°C × 20–30 min) – Drain at 80°C
Wetting Agent	Zetawet SWLF CONC	0.4%–0.6%	
Sequestering Agent	Zetaplex SQT CONC	0.3%–0.5%	
Peroxide Stabilizer	ZETASTAB BR	0.2%–0.3%	
Alkali	Caustic Soda Flakes	1.2%–2.5%	
Oxidative Bleaching Agent	Hydrogen Peroxide 50%	1.5%–3.0%	

2. Scouring & Oxidative Bleaching Bath — Full White (FW) Fabrics

Function	Product	Dosage	Parameters & Conditions
Lubricating Agent	Zetalub EAC	0.5%–1.0%	Auxiliary dosing – raise to 60°C – Caustic Soda (2 cycles) – H ₂ O ₂ 50% (2 cycles) – raise to optimum temp & hold – drain at 80°C – Hot wash at 80°C, add OBA (20 min dosing + 20 min holding) – Drain bath
Wetting Agent	Zetawet SWLF CONC	0.4%–0.6%	
Sequestering Agent	Zetaplex SQT CONC	0.3%–0.5%	
Peroxide Stabilizer	ZETASTAB BR	1.0%–1.5%	
Alkali	Caustic Soda Flakes	2.0%–3.0%	
Oxidative Bleaching Agent	Hydrogen Peroxide 50%	6.0%–8.0%	
OBA (base)	Zetabright HCE	0.7%–1.0%	
OBA (tint)	Zetabright BB	0.02%–0.1%	

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